Appendix B: PRATT Preoperative Investigation Recommendations

Preoperative testing decisions are typically guided by surgical invasiveness, anticipated blood loss, and patient conditions or medications. The goal was to develop an algorithm that uses PRATT patient screening responses and surgical invasiveness input to generate a tailored set of preoperative investigations aligned with relevant clinical guidelines. These recommendations are made available to clinicians early in the presurgical journey to guide evidence-based ordering of investigations.

Algorithm Development Process

In the initial development phase, the PCADS committee compiled all preoperative investigation tools across BC and refined them into a single set of recommendations. Relevant guidelines, including Choosing Wisely (1) and the CAS 2024 Guidelines to the Practice of Anesthesia (2), were incorporated. The PRATT tool was designed to differentiate between symptomatic and asymptomatic patients, allowing recommendations for 'no preoperative investigations' for asymptomatic patients undergoing low-risk surgery.

Surgery Classification based on Invasiveness/Stress or Risk

To support the algorithm, the Committee selected a published tool to classify surgeries by invasiveness and cross-referenced it with BC surgical codes. Surgical risk assessment tools like the NSQIP Calculator, SORT tool, and ACC/AHA guidelines are limited in distinguishing surgical-specific risks, as they calculate overall rather than surgery-specific risks. The Operative Stress Score (OSS) provides a more effective framework for stratifying preoperative consultations and investigations by categorizing surgeries into five stress levels (4). Using a Delphi method, a panel of surgeons and anesthesiologists classified common procedures into these levels, which the PCADS committee further streamlined into surgical invasiveness ranking (low, moderate, and high) to align with BC's most common procedures. This adaptation will support a searchable, clinically applicable tool. Examples include:

Low or Very Low Surgical Stress (OSS 1 or 2)	Moderate to High Surgical Stress (OSS 3 or 4)	Very High Surgical Stress (OSS 5)
● ENT: microlaryngoscopy, septorhinoplasty, otology, tonsillectomy, thyroidectomy ● Dental: restorations/extractions ● Plastics: extremity surgery ● Spine: lumbar discectomy ● Urol: cystoscopy ● GenSx: lymph node biopsy, simple hernia, laparoscopic appendectomy, partial/simple mastectomy	● ENT: radical neck dissection, major head and neck, laryngectomy ● Plastics: breast flap reconstruction ● Spine: laminectomy, instrumentation, fusion more than 2 levels ● Urol: laparoscopic prostatectomy, adrenalectomy, TURP, TURBT ● GenSx: laparoscopic bowel	● Cardiac surgery ● Thoracics: lung resection, esophageal resection ● Vascular: open aortic surgery, lower extremity open revascularization ● Neuro: aneurysm/AVM, skull base tumor ● Ortho: complex joint revision ● General surgery: cytoreductive surgery/HIPEC, multivisceral resection, pelvic

● Optho: cataract ● Ortho: hand, foot, arthroscopic shoulder, ankle ● GYNE: hysteroscopy, D&C, laparoscopic salpingectomy, ● Endoscopic and superficial procedures	resections, laparascopic chole, radical mastectomy, laparoscopic Nissen, gastric restrictive procedures ● Ortho: total arthroplasty hip, knee, shoulder ● Neuro: convexity/superficial tumor, epilepsy, deep brain stimulation ● Gyne: laparoscopic or open hysterectomy, myomectomy ● Vasc: carotid endarterectomy	exenteration, whipple ● Urology: Radical Cystectomy, Nephrectomy (partial or total)

Preoperative Testing Recommendations based on Surgical Invasiveness Ranking

For low surgical invasiveness procedures, preoperative investigations are minimal, with no tests for asymptomatic patients or a grid-guided approach based on specific symptoms if present. For moderate surgical invasiveness procedures, a baseline complete blood count (CBC) is recommended to detect and manage potential anemia, and renal function tests are suggested to assess NSAID suitability as co-analgesics. Additional tests for moderate procedures may be added based on patient-specific conditions, such as hypertension. For high surgical invasiveness procedures an HbA1c is recommended to identify and manage undiagnosed diabetes.

Identifying Asymptomatic Patients

Evidence suggests that asymptomatic patients undergoing low-risk surgery do not benefit from routine tests, as abnormalities rarely impact perioperative management (1). A screening approach was developed to identify asymptomatic patients with or without coexisting conditions who may **not** require preoperative investigations. This assessment checks for active cardiorespiratory symptoms, poor functional capacity, frailty, and any history of bleeding disorders, anemia, active bleeding, or renal dysfunction. Patients without these indicators are considered asymptomatic and if undergoing elective, low-stress surgeries, no preoperative tests are recommended. However, for patients with poor functional capacity, frailty, or active symptoms, a structured grid provides guidance for necessary preoperative investigations.

The following variables are used to define asymptomatic patients for the purposes of preoperative testing:

- *DASI>34*
- *FRAIL SCALE <3*
- *Not admitted to hospital in the last year*
- *No shortness of breath at rest, walking slowly on flat ground, or progressing over the past few months*
- *Breathing back to normal after recent chest infection*

- *No difficulty lying flat with just 1 pillow because of breathing problems*
- *No oxygen use at home*
- *No current chest pain, pressure or tightness (i.e., angina or heart pain)*
- *No fainting spells, passing out or syncope with exercise*
- *No current anemia or low blood counts*
- *No bleeding disorder such as hemophilia or Von Willebrand's Disease*
- *No active or ongoing bleeding within the past 12 months*
- *No low or poor kidney function*

Preoperative Testing Based on Patient Comorbid Disease

For patients undergoing moderate or high OSS procedures, their medications and comorbidities help to guide preoperative investigations. The Perioperative Investigations Recommendation grid (Appendix B2) outlines the recommended investigations based on patient comorbidities, along with supporting quotes from perioperative or primary care guidelines where applicable. If no guideline is referenced, the clinical reasoning behind the recommendation is provided in the final column.

Incorporation into PRATT

The PRATT algorithm integrates patient responses, OSS score (entered by PAC/MOA), and hospital stay status to generate tailored preoperative investigation recommendations. For **low surgical invasiveness procedures**, asymptomatic patients do not require preoperative investigations, while symptomatic patients receive condition-specific recommendations. For **moderate surgical invasiveness procedures**, baseline tests (CBC, Cr, and eGFR) are conducted, with additional tests as indicated by PRATT responses and risk factors. For **high surgical invasiveness procedures**, baseline tests (CBC, Cr, eGFR, and EKG) are standard, with further investigations based on PRATT responses and identified risks.

The final preoperative investigation recommendations are included in the clinical output summary. This information can be used to generate a laboratory requisition, which could be sent to the patient after review and validation by a clinician (i.e., PAC nurse).

Additional PCADS Recommendations

- Timing of Preoperative Investigations:
    - Preoperative investigations should not "expire" automatically (2).
    - A timeline of within 6 months of the surgical date is suggested if the patient has not had any new medical diagnoses, no new medications started, and no admissions to hospital in this timeframe (2).
    - Do not reorder preoperative tests that have been completed within this timeframe if the test was within normal range and patient health status is unchanged. (e.g., if the test has already been completed for other reasons)
    - Should 6 months expire, or the patient has a new medical condition, new medication, or admission to hospital, anesthesiology review is suggested to advise on what laboratory investigations need to be repeated.
- Preoperative Group and Screen:

  - Institutional guidelines are recommended to determine which surgical procedures require group and screen prior to surgery.
  - These guidelines should be available through the transfusion medicine service and are reflective of an institutional transfusion rate of >1%.
  - If a patient requires a preoperative group & screen, a preoperative CBC is also recommended.
- HbA1c:
  - For patients with known diabetes a preoperative HbA1c within the last 6 months is appropriate to understand blood glucose control and to guide preoperative optimization is necessary.
  - Based on the Diabetes Canada screening recommendations, patients having high surgical invasiveness surgeries, and patients with baseline risk factors undergoing moderate surgical invasiveness surgeries, should have preoperative HbA1c testing (5).

References

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Primary Care Screening Guidelines
Perioperative Care Screening Guidelines

Patient Coexisting Disease (Recommend in mod/high OSS procedures only)	CBC	Na K	eGFR Cr	HgbA1c	INR	Albumin	ALT ALP Bilirubin	GGT	NT-proBNP	EKG	TSH	CXR	Lithium Level	Digoxin Level	HCO3	PTT	References and/or Clinical Reasoning
Age > 70	x		x	x													1. CAS Guidelines 2024 (3): CBC Extremes of age May be considered based on the potential for significant blood loss, extremes of age, liver or hematological disease, history of anemia or malignancy. 2. CAS Guidelines 2024 (3): Creatinine and estimated GFR Advanced age and for patients taking medications potentially affecting renal function • for patients receiving direct oral anticoagulants • as required for calculating perioperative risk indices • eGFR is recommended to assist with renal outcome prediction. 3. ASA Practice Advisory for Preanesthesia Evaluation 2012 (2): The Task Force recognizes that age alone may not be an indication for ECG
Age >= 40				x													Canadian CPG Diabetes 2018 (5): If you are age 40 years or over, you are at risk for type 2 diabetes and should be tested at least every 3 years. • If you have risk factors that increase the likelihood of developing type 2 diabetes, you should be tested more frequently and/or start regular screening earlier. Some of the risk factors include family history of diabetes; being a member of a high-risk population; history of prediabetes or gestational diabetes; and being overweight.
Hypertension		x	x	x													CAS Guidelines 2024 (3): Electrolytes & Creatinine and estimated glomerular filtration rate (eGFR) May be considered: • with known or compelling findings in favour of hypertension, congestive heart failure, chronic renal failure, complicated diabetes, liver disease, pituitary or adrenal disease, or malnutrition • for patients taking diuretics, angiotensin-converting enzyme inhibitors, angiotensin receptor blockers, and other therapy affecting electrolytes. CKD Pathway (6) Testing for CKD should not be universal, but should be targeted for individuals at increased risk of developing CKD: Hypertension Diabetes Mellitus Family history of Stage 5 CKD or hereditary kidney disease Vascular disease (prior diagnosis of CVD, stroke/TIA or PVD) Multisystem disease with potential kidney involvement (e.g. SLE) Canadian Diabetes Association (5) recommends screening every 3 years.

Patient Coexisting Disease (Recommend in mod/high OSS procedures only)	CBC	Na K	eGFR Cr	HgbA1c	INR	Albumin	ALT ALP Bilirubin	GGT	NT-proBNP	EKG	TSH	CXR	Lithium Level	Digoxin Level	HCO3	PTT	References and/or Clinical Reasoning
Congestive Heart Failure		x	x						x								CAS Guidelines 2024 (3): Electrolytes & Creatinine and estimated glomerular filtration rate (eGFR) May be considered: • with known or compelling findings in favour of hypertension, congestive heart failure, chronic renal failure, complicated diabetes, liver disease, pituitary or adrenal disease, or malnutrition • for patients taking diuretics, angiotensin-converting enzyme inhibitors, angiotensin receptor blockers, and other therapyaffecting electrolytes.
History of ischemic heart disease **Procedures such as heart stents, open heart surgery (CABG or bypass surgery), or heart valve surgery**			x	x						x							CAS Guidelines 2024 (3): Electrocardiography May be considered for patients with known or suspected coronary heart disease, significant arrhythmia, peripheral vascular disease, or other significant structural heart disease • May be considered in the absence of symptoms or known cardiovascular disease in patients having high-risk surgery in the presence of clinical risk factors (e.g., Revised Cardiac Risk Index or American College of Surgeons Surgical Risk Calculator). CKD Pathway (6) Testing for CKD should not be universal, but should be targeted for individuals at increased risk of developing CKD: Hypertension Diabetes Mellitus Family history of Stage 5 CKD or hereditary kidney disease Vascular disease (prior diagnosis of CVD, stroke/TIA or PVD) Multisystem disease with potential kidney involvement (e.g. SLE) Canadian Diabetes Association (5) recommends screening HbA1c every 3 years.
History of Atrial fibrillation **- Heart rhythm disorders or fast heart rate syndromes (like intermittent SVT, supraventricular tachycardia, Brugada syndrome, intermittent ventricular tachycardia, congenital long QT)** **- CIED**										x							CAS Guidelines 2024 (3): Electrocardiography May be considered for patients with known or suspected coronary heart disease, significant arrhythmia, peripheral vascular disease, or other significant structural heart disease • May be considered in the absence of symptoms or known cardiovascular disease in patients having high-risk surgery in the presence of clinical risk factors (e.g., Revised Cardiac Risk Index or American College of Surgeons Surgical Risk Calculator).

Patient Coexisting Disease (Recommend in mod/high OSS procedures only)	CBC	Na K	eGFR Cr	HgbA1c	INR	Albumin	ALT ALP Bilirubin	GGT	NT-proBNP	EKG	TSH	CXR	Lithium Level	Digoxin Level	HCO3	PTT	References and/or Clinical Reasoning
History of Valvular Disease										x							CAS Guidelines 2024 (3): Electrocardiography May be considered for patients with known or suspected coronary heart disease, significant arrhythmia, peripheral vascular disease, or other significant structural heart disease • May be considered in the absence of symptoms or known cardiovascular disease in patients having high-risk surgery in the presence of clinical risk factors (e.g., Revised Cardiac Risk Index or American College of Surgeons Surgical Risk Calculator).
Peripheral Vascular Disease			x	x						x							CAS Guidelines 2024 (3): electrocardiography May be considered for patients with known or suspected coronary heart disease, significant arrhythmia, peripheral vascular disease, or other significant structural heart disease • May be considered in the absence of symptoms or known cardiovascular disease in patients having high-risk surgery in the presence of clinical risk factors (e.g., Revised Cardiac Risk Index or American College of Surgeons Surgical Risk Calculator). CKD Pathway (6) Testing for CKD should not be universal, but should be targeted for individuals at increased risk of developing CKD: Hypertension Diabetes Mellitus Family history of Stage 5 CKD or hereditary kidney disease Vascular disease (prior diagnosis of CVD, stroke/TIA or PVD) Multisystem disease with potential kidney involvement (e.g. SLE) Canadian Diabetes Association (5) recommends screening HbA1c every 3 years.
History of Stroke/TIA			x	x						x							CAS Guidelines 2024 (3): electrocardiography May be considered for patients with known or suspected coronary heart disease, significant arrhythmia, peripheral vascular disease, or other significant structural heart disease • May be considered in the absence of symptoms or known cardiovascular disease in patients having high-risk surgery in the presence of clinical risk factors (e.g., Revised Cardiac Risk Index or American College of Surgeons Surgical Risk Calculator). CKD Pathway (6) Testing for CKD should not be universal, but should be targeted for individuals at increased risk of developing CKD: Hypertension Diabetes Mellitus Family history of Stage 5 CKD or hereditary kidney disease Vascular disease (prior diagnosis of CVD, stroke/TIA or PVD) Multisystem disease with potential kidney involvement (e.g. SLE) Canadian Diabetes Association (5) recommends screening HbA1c every 3 years.

Patient Coexisting Disease (Recommend in mod/high OSS procedures only)	CBC	Na K	eGFR Cr	HgbA1c	INR	Albumin	ALT ALP Bilirubin	GGT	NT-proBNP	EKG	TSH	CXR	Lithium Level	Digoxin Level	HCO3	PTT	References and/or Clinical Reasoning
Cystic Fibrosis				x							x						CAS Guidelines 2024 (3): Chest radiograph May be considered for patients with acute or chronic cardiopulmonary disease based on history and physical exam if it will change management. Canadian Diabetes Association (5) recommends screening HbA1c every 3 years
Blood Diseases (i.e., hemochromatosis, sickle cell disease, myelodysplasia, chronic leukemia	x																CAS Guidelines 2024 (3): CBC May be considered based on the potential for significant blood loss, extremes of age, liver or hematological disease, history of anemia or malignancy.
Diabetes (type 1 or 2 or hx GDM or pre-diabetes)		x	x	x													Follow HbA1c every 6 months, check GFR as renal disease commonly associated. CAS Guidelines 2024 (3): Electrolytes & Creatinine and estimated glomerular filtration rate (eGFR) May be considered: • with known or compelling findings in favour of hypertension, congestive heart failure, chronic renal failure, complicated diabetes, liver disease, pituitary adrenal disease, or malnutrition • for patients taking diuretics, angiotensin-converting enzyme inhibitors, angiotensin receptor blockers, and other therapy affecting electrolytes. CAS Guidelines 2024 (3): HbA1c May be considered for patients with known diabetes mellitus as early as possible before surgery (ideally at time of surgical referral) if results would change management. CPG CJD 2018 (5): HbA1c If you are age 40 years or over, you are at risk for type 2 diabetes and should be tested at least every 3 years. • If you have risk factors that increase the likelihood of developing type 2 diabetes, you should be tested more frequently and/or start regular screening earlier. Some of the risk factors include family history of diabetes; being a member of ahigh-risk population; history of prediabetes or gestational diabetes; and having overweight. CKD Pathway (6) Testing for CKD should not be universal, but should be targeted for individuals at increased risk of developing CKD: Hypertension Diabetes Mellitus Family history of Stage 5 CKD or hereditary kidney disease Vascular disease (prior diagnosis of CVD, stroke/TIA or PVD) Multisystem disease with potential kidney involvement (e.g. SLE)

Patient Coexisting Disease (Recommend in mod/high OSS procedures only)	CBC	Na K	eGFR Cr	HgbA1c	INR	Albumin	ALT ALP Bilirubin	GGT	NT-proBNP	EKG	TSH	CXR	Lithium Level	Digoxin Level	HCO3	PTT	References and/or Clinical Reasoning
Rheumatological disease			x														Risks of anemia and renal dysfunction. CKD Pathway (6) screening guidelines CKD Pathway (6) Testing for CKD should not be universal, but should be targeted for individuals at increased risk of developing CKD: Hypertension Diabetes Mellitus Family history of Stage 5 CKD or hereditary kidney disease Vascular disease (prior diagnosis of CVD, stroke/TIA or PVD) Multisystem disease with potential kidney involvement (e.g. SLE)
Thyroid disease (hypo or hyperthyroidism with unknown TSH value)											x						May need optimization
Malnutrition (CNST 2)	x		x		x												CAS Guidelines 2024 (3):INR/PT with conditions or medications associated with impaired coagulation (e.g., liver disease, malnutrition), history of excessive bleeding, or a family history of heritable coagulopathies • for patients on oral or parenteral anticoagulant therapy. CAS Guidelines 2024 (3): Electrolytes & Creatinine and estimated glomerular filtration rate (eGFR) May be considered: • with known or compelling findings in favour of hypertension, congestive heart failure, chronic renal failure, complicated diabetes, liver disease, pituitary or adrenal disease, or malnutrition • for patients taking diuretics, angiotensin-converting enzyme inhibitors, angiotensin receptor blockers, and other therapyaffecting electrolytes.
Diuretics (drug library)		x	x														CAS Guidelines 2024 (3): Electrolytes & Creatinine and estimated glomerular filtration rate (eGFR) May be considered: • with known or compelling findings in favour of hypertension, congestive heart failure, chronic renal failure, complicated diabetes, liver disease, pituitary or adrenal disease, or malnutrition • for patients taking diuretics, angiotensin-converting enzyme inhibitors, angiotensin receptor blockers, and other therapyaffecting electrolytes.
ACEi/ARBs (drug library)		x	x														CAS Guidelines 2024 (3): Electrolytes & Creatinine and estimated glomerular filtration rate (eGFR) May be considered: • with known or compelling findings in favour of hypertension, congestive heart failure, chronic renal failure, complicated diabetes, liver disease, pituitary or adrenal disease, or malnutrition • for patients taking diuretics, angiotensin-converting enzyme inhibitors, angiotensin receptor blockers, and other therapyaffecting electrolytes.

Patient Coexisting Disease (Recommend in mod/high OSS procedures only)	CBC	Na K	eGFR Cr	HgbA1c	INR	Albumin	ALT ALP Bilirubin	GGT	NT-proBNP	EKG	TSH	CXR	Lithium Level	Digoxin Level	HCO3	PTT	References and/or Clinical Reasoning
Warfarin (drug library)	x				x												CAS Guidelines 2024 (3): INR/PT with conditions or medications associated with impaired coagulation (e.g., liver disease, malnutrition), history of excessive bleeding, or a family history of heritable coagulopathies • for patients on oral or parenteral anticoagulant therapy.
Lithium use (drug library)			x								x		x				Preoperative baseline
SLGT-2i (drug library)															x		For baseline in event of euglycemic diabetic ketoacidosis
Steroid use (drug library)				x													Canadian Diabetes Association (5) recommends screening every 3 years CAS 2024 Guidelines (3): Fasting Glucose Level (but used A1c due to no need for fasting) May be considered: • for patients with diabetes mellitus, preoperatively on day of surgery to guide glycemic control • for patients on glucocorticoid therapy • as screening for body mass index >= 40 kg/m2 or a very high risk of diabetes mellitus based on signs and symptoms.
Digoxin use (drug library)			x											x			
DOAC (drug library)	x		x														CAS Guidelines 2024 (3): Creatinine and estimated glomerular filtration rate (eGFR): for patients receiving direct oral anticoagulants • as required for calculating perioperative risk indices • eGFR is recommended to assist with renal outcome prediction.
BMI >=_40 (demographics)				x											x		CAS Guidelines 2024 (3): Fasting Glucose Level (but used A1c due to no need for fasting) May be considered: • for patients with diabetes mellitus, preoperatively on day of surgery to guide glycemic control • for patients on glucocorticoid therapy • as screening for body mass index >= 40 kg/m2 or a very high risk of diabetes mellitus based on signs and symptoms. Canadian Diabetes Association (5): HbA1c If you are age 40 years or over, you are at risk for type 2 diabetes and should be tested at least every 3 years. • If you have risk factors that increase the likelihood of developing type 2 diabetes, you should be tested more frequently and/or start regular screening earlier. Some of the risk factors include family history of diabetes; being a memberof a high-risk population; history of prediabetes or gestational diabetes; and having overweight.
Sleep Disordered Breathing: - dx OSA - STOP-BANG >= 5 - Other breathing disorder (OHVS)				x						x					x		Markers of disease severity: elevated HCO3 and evidence of right heart strain Canadian Diabetes Association (5) recommends screening HbA1c, polycythemia and HCO3 if there is a clinical suspicion of sleep disordered breathing.
Urologic Surgery (Surgery type)		x	x														High risk of coexisting renal disease, may need for dosing commonly used antibiotics such as gentamycin.
DASI Flag <34	x		x							x							Patient has a low cardiorespiratory fitness according to DASI screening or has some cardiorespiratory symptom.
FRAIL SCALE Flag >=3	x		x							x							Positive Frailty Screening.
Hospitalized within the last 6 months	x		x							x							Potential unstable medical condition.

Patient Coexisting Disease (Recommend in mod/high OSS procedures only)	CBC	Na K	eGFR Cr	HgbA1c	INR	Albumin	ALT ALP Bilirubin	GGT	NT-proBNP	EKG	TSH	CXR	Lithium Level	Digoxin Level	HCO3	PTT	References and/or Clinical Reasoning
Active Respiratory Symptoms: **Shortness of breath:** **- at rest** **- walking slowly on flat ground** **- progressing over the past few months** **Home Oxygen** **Unable to lay flat due to breathing difficulties** **Breathing not back to normal after recent chest infection**	x		x							x		x					Symptomatic cardiorespiratory disease. CAS Guidelines 2024 (3): Chest radiograph may be considered for patients with acute or chronic cardiopulmonary disease based on history and physical exam *if it will change management* .
Active Cardiac Symptoms: **Experiencing chest pain, pressure or tightness at rest, new in the last few weeks or months, coming on with less activity or taking longer to go away** **Experience fainting spells, passing out or syncope with exercise, within the last month, occurring daily/weekly/monthly**	x		x							x							
RCRI >=1 or AGE ≥65 **+** **Inpatient stay postoperatively**									x	x							CCS 2017 guidelines (7) CAS Guidelines 2024 (3): NT proBNP May be considered, when hospital admission is anticipated, if: • Age[65 yr • Age 45–65 yr with significant cardiac disease • Revised Cardiac Risk Index score >= 1.

Patient Coexisting Disease (Recommend in mod/high OSS procedures only)	CBC	Na K	eGFR Cr	HgbA1c	INR	Albumin	ALT ALP Bilirubin	GGT	NT-proBNP	EKG	TSH	CXR	Lithium Level	Digoxin Level	HCO3	PTT	References and/or Clinical Reasoning
Known Cirrhosis, Hepatitis C, Hepatitis B **EtOH excess AUDIT-C >= 5**	x	x	x		x	x	x	x								x	BC Guidelines abnormal liver chemisty CAS Guidelines 2024 (3): INR/PT with conditions or medications associated with impaired coagulation (e.g., liver disease, malnutrition), history of excessive bleeding, or a family history of heritable coagulopathies • for patients on oral or parenteral anticoagulant therapy. AUDIT-C >= 5 suggests: High risk of AUD and risk of liver dysfunction Child-Pugh Classification/ Vocal-Penn/ MELD-Na require these for risk stratification prior to surgery.
Chronic Kidney Disease	x	x	x												x		CAS Guidelines 2024 (3): Electrolytes & Creatinine and estimated glomerular filtration rate (eGFR) May be considered: • with known or compelling findings in favour of hypertension, congestive heart failure, chronic renal failure, complicated diabetes, liver disease, pituitary or adrenal disease, or malnutrition • for patients taking diuretics, angiotensin-converting enzyme inhibitors, angiotensin receptor blockers, and other therapy affecting electrolytes.
Current Anemia **Blood or iron infusions in the last 6 months** **Experiencing active ongoing bleeding such as blood in urine, stool or very heavy vaginal bleeding**	x																CAS Guidelines 2024 (3): CBC May be considered based on the potential for significant blood loss, extremes of age, liver or hematological disease, history of anemia or malignancy.
Current Cancer **Chemo or radiation in last 1 month**	x																CAS Guidelines 2024 (3): CBC May be considered based on the potential for significant blood loss, extremes of age, liver or hematological disease, history of anemia or malignancy.
Bleeding Disorder	x				x											x	CAS Guidelines 2024 (3): INR/PT with conditions or medications associated with impaired coagulation (e.g., liver disease, malnutrition), history of excessive bleeding, or a family history of heritable coagulopathies • for patients on oral or parenteral anticoagulant therapy.

Patient Coexisting Disease (Recommend in mod/high OSS procedures only)	CBC	Na K	eGFR Cr	HgbA1c	INR	Albumin	ALT ALP Bilirubin	GGT	NT-proBNP	EKG	TSH	CXR	Lithium Level	Digoxin Level	HCO3	PTT	References and/or Clinical Reasoning
Symptomatic Patient Questionss: (no to any symptom related questions - see below)	No Preoperative Testing in Low Invasiveness Surgery																CAS Guidelines 2024 (3): Routine laboratory blood testing, electrocardiograms, and chest radiographs are not recommended for asymptomatic patients having low-risk surgery. Examples of low-risk surgery include, but are not restricted to, cataract surgery, minor breast surgery, superficial procedures, and many minor ambulatory procedures. Choosing Wisely Canada 2023 (1): Don't order baseline laboratory studies (complete blood count, coagulation testing, or serum biochemistry) for asymptomatic patients undergoing low-risk non-cardiac surgery. Conducting baseline laboratory investigations before low-risk non-cardiac surgery contributes little value to perioperative care. A focused clinical history and physical examination may reliably identify relevant abnormalities sought by routine laboratory testing before low-risk surgery. In addition, evidence suggests that abnormal results in this setting only rarely influence management and do not improve clinical outcomes. Preoperative testing may add value in the setting of a symptomatic patient or higher-risk surgery, but should not be performed routinely before low-risk surgery on asymptomatic patients.
Very High Surgical Stress (OSS 5):	x		x	x						x							Baseline Orders: - CBC (optimize anemia if present) - eGFR (baseline and safety of NSAID as coanalgesic postop) - HbA1c (potential benefit for catching untreated diabetes and initiating therapy prior to very high surgical stress procedures) + Grid Output for Patient Comorbidities Impact CAS Guidelines 2024 (3): Creatinine and estimated GFR Advanced age and for patients taking medications potentially affecting renal function • for patients receiving direct oral anticoagulants • as required for calculating perioperative risk indices • eGFR is recommended to assist with renal outcome prediction. CAS Guidelines 2024 (3): Electrocardiography May be considered for patients with known or suspected coronary heart disease, significant arrhythmia, peripheral vascular disease, or other significant structural heart disease • May be considered in the absence of symptoms or known cardiovascular disease in patients having high-risk surgery in the presence of clinical risk factors (e.g., Revised Cardiac Risk Index or American College of Surgeons Surgical Risk Calculator).

Patient Coexisting Disease (Recommend in mod/high OSS procedures only)	CBC	Na K	eGFR Cr	HgbA1c	INR	Albumin	ALT ALP Bilirubin	GGT	NT-proBNP	EKG	TSH	CXR	Lithium Level	Digoxin Level	HCO3	PTT	References and/or Clinical Reasoning
Moderate to High Surgical Stress (OSS 3 or 4):	x		x														Baseline Orders: - CBC (optimize anemia if present) - GFR (baseline and safety of NSAID as coanalgesic postop) + Grid Output for Patient Comorbidities Impact CAS Guidelines 2024 (3): Creatinine and estimated GFR Advanced age and for patients taking medications potentially affecting renal function • for patients receiving direct oral anticoagulants • as required for calculating perioperative risk indices • eGFR is recommended to assist with renal outcome prediction.
Low or Very Low Surgical Stress (OSS 1 or 2):	No Preop Investigations if Asymptomatic																Asymptomatic means: - DASI>34 - FRAIL SCALE <3 - No to "admitted to hospital in the last year" - No to: "shortness of breath at rest and shortness of breath walking slowly on the flat and shortness of breath progressing over the past few months" - No to "breathing not back to normal after recent chest infection" - No to "have difficulty laying flat (with just 1 pillow) because of breathing problems"? - No to "use oxygen at home" - No to "currently experience chest pain, pressure or tightness (also called angina or heart pain)" - No to "had fainting spells, passing out or syncope with exercise, within the last month, occurring daily/weekly/monthly, have not seen a doctor about it - No to "current anemia or low blood counts" - No to "have a bleeding disorder such as hemophilia or Von Willebrand's Disease" - No to "you experienced active or ongoing bleeding within the past 12 months (e.g., blood in your urine, blood in your stool, heavy vaginal bleeding, etc.)? - No to "low or poor kidney function (also called chronic kidney disease or kidney failure)"?

Provincial Procedure Group Name	Provincial Procedure Code	Provincial Procedure Name	Surgical Invasiveness Ranking	Low Risk Urologic Surgery Needing Preop GFR Na K
Other Eye Surgery	11001	2ndy Iol Implant/Repositioning Implant Add/Remove	low	
Abdominoplasty	80001	Abdominoplasty	moderate	
Tonsillectomy/Adenoidectomy	50001	Adenoidectomy	low	
All Other Procedures	60001	Adrenalectomy	moderate	
Oral Cavity & Pharynx Surgery	90009	Alveoplasty/Bone Recontouring	low	
Other Orthopaedic Surgery	10144	Amputation	moderate	
All Other Procedures	30001	Amputation Cervix	moderate	
Rectal Surgery	20040	Anal Fissurectomy/Fistulectomy	low	
Vascular Surgery - Other	40034	Angiography	low	
Not Reported	40033	Angioplasty	low	
Rectal Surgery	20002	Anoplasty	low	
Lens & Vitreous (non-cataract) Surgery	11002	Anterior Vitrectomy /Lensectomy	low	
Shunt Insertion/Removal	50002	Antrostomies	low	
Not Reported	12010	Aortic	high	
Appendectomy	20003	Appendectomy	moderate	
Other Orthopaedic Surgery	10012	Application Cast	low	
Spinal/Back Surgery	70041	Application Cervical Traction	moderate	
Oral Cavity & Pharynx Surgery	90008	Arch Bar Application	low	
Other Joint Reconstruction	10146	Arthrodesis/Fusion/Reconstruction	moderate	
Foot/Ankle Surgery	10024	Arthroplasty Ankle Replacement	moderate	
Foot/Ankle Surgery	10162	Arthroplasty Ankle Revision	moderate	
Other Joint Reconstruction	10025	Arthroplasty Elbow Replacement	moderate	
Other Joint Reconstruction	10165	Arthroplasty Elbow Replacement Revision	moderate	
Foot/Ankle Surgery	10026	Arthroplasty Foot/Toe Replacement	low	
Other Joint Reconstruction	10028	Arthroplasty Hip Hemi	moderate	
Hip Replacement	10029	Arthroplasty Hip Revision	moderate	
Knee Replacement	10032	Arthroplasty Knee Replacement	low	
Knee Replacement	10031	Arthroplasty Knee Revision	moderate	
Shoulder Surgery	10033	Arthroplasty Shoulder Replacement	moderate	
Shoulder Surgery	10166	Arthroplasty Shoulder Replacement Revision	moderate	
Foot/Ankle Surgery	10036	Arthroscopy Ankle	low	
Other Joint Reconstruction	10037	Arthroscopy Elbow	low	
Foot/Ankle Surgery	10038	Arthroscopy Foot	low	
Other Joint Reconstruction	10039	Arthroscopy Hip	low	
Knee - Meniscectomy	10040	Arthroscopy Knee	low	
Facial Bone Reconstruction	90021	Arthroscopy Of The Temporomandibular Joint	low	
Shoulder Surgery	10041	Arthroscopy Shoulder	low	
Other Orthopaedic Surgery	10147	Arthrotomy	low	
Facial Bone Reconstruction	90023	Arthrotomy/Arthroplasty Of The Temporomandibular Joint	low	
Cyst/Ganglion Removal	10157	Aspiration/Excision Cyst Or Ganglion	low	

Provincial Procedure Group Name	Provincial Procedure Code	Provincial Procedure Name	Surgical Invasiveness Ranking	Low Risk Urologic Surgery Needing Preop GFR Na K
Vascular Surgery - Other	40001	Av Fistula Formation	low	
Biopsy in OR	10051	Biopsy Bone	low	
Cranial Surgery	70001	Biopsy Brain	low	
Biopsy in OR	20112	Biopsy Mucous Membrane	low	
Biopsy in OR	10052	Biopsy Muscle	low	
Biopsy in OR	70002	Biopsy Nerve	low	
Prostate Surgery	60005	Biopsy Prostate	low	
Biopsy in OR	10053	Biopsy Synovial	low	
Biopsy in OR	60007	Biopsy Testicle	low	
Biopsy in OR	20111	Biopsy Visceral	low	
Biopsy in OR	20147	Biopsy/Dissection Lymph Node	low	
Other Urology Surgery	60068	Bladder/ Urethral Diverticulectomy	low	
Other Eye Surgery	11003	Blepharoplasty	low	
Other Orthopaedic Surgery	10054	Bone Graft	moderate	
All Other Procedures	20114	Brachytherapy	moderate	
All Other Procedures	80099	Breast Augmentation - Transgender	low	
Not Reported	100004	Bronchoscopy With Or Without Biopsy	low	
Foot/Ankle Surgery	10055	Bunionectomy	low	
Cranial Surgery	70003	Burr Holes	moderate	
Not Reported	12003	Cabg + Other	high	
Not Reported	12002	Cabg + Valve	high	
Not Reported	12001	Cabg + Valve + Other	high	
Uterine Surgery	30007	Caesarean Section	moderate	
Other Eye Surgery	11004	Canthoplasty/Pexy	low	
Breast Reconstruction	80003	Capsulotomy/Capsulectomy Breast	low	
Cataract Surgery	11005	Cataract Extraction & Iol Implant	low	
Not Reported	12016	Cath Lab Procedure	moderate	
Nasal Surgery	50012	Cautery Nasal Septum	low	
All Other Procedures	30032	Cervical Cerclage Suture	low	
Cone Biopsy	30009	Cervical Conization	low	
Oral Cavity & Pharynx Surgery	80004	Cheiloplasty	low	
Cholecystectomy	20010	Cholecystectomy	moderate	
All Other Procedures	20011	Cholecystojejunostomy	moderate	
Male Reproductive Surgery	60009	Circumcision	low	
All Other Procedures	20047	Closure Of Gastrostomy	low	
Nasal Surgery	50014	Closure Septal Perforation	low	
Spinal/Back Surgery	10072	Coccygectomy	low	
Other Ear Surgery	50062	Cochlear Implant	moderate	
Colostomy/Ileostomy	20017	Colostomy	moderate	
Colostomy/Ileostomy	20019	Colostomy Reversal	moderate	
All Other Procedures	30010	Colpectomy	low	
All Other Procedures	30012	Colposcopy	low	
All Other Procedures	30013	Colpotomy	low	

Provincial Procedure Group Name	Provincial Procedure Code	Provincial Procedure Name	Surgical Invasiveness Ranking	Low Risk Urologic Surgery Needing Preop GFR Na K
Wound/Laceration Care	20115	Complex Dressing Change	low	
Not Reported	12013	Congenital Heart	high	
All Other Procedures	60080	Congenital Urology/Gynecology Surgery	low	
Other Urology Surgery	60010	Continent Urinary Diversion	moderate	
Wound/Laceration Care	20136	Control Hemorrhage	low	
Not Reported	11048	Cornea Transplant	low	
Cranial Surgery	70007	Craniectomy/Craniotomy	high	
Cranial Surgery	70008	Cranioplasty	low	
Other Eye Surgery	11008	Cryopexy	low	
Other Eye Surgery	11009	Cycloablative Procedures	low	
Bladder Surgery	60011	Cystectomy- Partial Or Radical, With Or Without Ileal Conduit	moderate	
Kidney/Bladder Stone Removal	60012	Cystolithopaxy/Cystolithotomy	low	
Other Urology Surgery	60014	Cystoscopy With Or Without Retrograde Pyelography	low	Y
Bladder Surgery	60015	Cystostomy	low	
Other Eye Surgery	11010	Dacrocystorhinostomy/Dacrocystostomy	low	
Spinal/Back Surgery	70009	Decompression Spinal	moderate	
Nerve Surgery	10073	Decompression/Lysis/Release Peripheral Nerve	low	
Lung Surgery	100005	Decortication Lung	moderate	
Wound/Laceration Care	80091	Delayed Primary Closure	low	
Dental Surgery	90010	Dental Cleaning	low	
Dental Surgery	90020	Dental Restorations	low	
Skin Surgery	80009	Dermabrasion	low	
Skin Surgery	80094	Dermis Fat Injection/Insertion	low	
All Other Procedures	20130	Diagnostic Imaging	low	
All Other Procedures	20127	Dilation	low	
D&C and Related Surgery	30016	Dilation And Curettage With Or Without Hysteroscopy Or Laparoscopy	low	
Spinal/Back Surgery	70010	Discectomy	moderate	
All Other Procedures	50015	Dissection Radical Neck	moderate	
Other Eye Surgery	11011	Duct Probing/Insertion Of Tube	low	
Other Eye Surgery	11012	Ectropion/Entropion/Epiplepharon	low	
Endarterectomy	40035	Endarterectomy	low	
Endarterectomy	40036	Endarterectomy Carotid	moderate	
D&C and Related Surgery	30017	Endometrial Ablation	low	
Sinus Surgery	50016	Endoscopic Sinus Surgery	moderate	
All Other Procedures	20153	Endoscopic Ultrasound	low	
Aortic Aneurysm Repair	40048	Endovascular Abdominal Aortic Aneurysm Repair	moderate	
Vascular Surgery - Other	40047	Endovascular Surgery	moderate	
Other Eye Surgery	11013	Enucleation/Evisceration - Orbital	low	
Not Reported	12015	Ep Procedure	moderate	
Male Reproductive Surgery	60018	Epididymectomy	low	
Other Orthopaedic Surgery	10164	Epiphysiodesis	low	
Esophagectomy	100024	Esophagectomy	high	
Esophagectomy	100008	Esophagogastrectomy	high	

Provincial Procedure Group Name	Provincial Procedure Code	Provincial Procedure Name	Surgical Invasiveness Ranking	Low Risk Urologic Surgery Needing Preop GFR Na K
Sinus Surgery	50018	Ethmoidectomy	low	
Wound/Laceration Care	20116	Evacuation Hematoma	low	
Other Eye Surgery	11014	Evacuation Of Hyphaema	low	
Examination Under Anaesthetic	20113	Examination/Exploration Under Anesthetic	low	
Excision Lesion/Tumour	30019	Excision Adnexal Mass	low	
Other Ear Surgery	50019	Excision Auricular Cyst	low	
Other Joint Reconstruction	10078	Excision Bursa	low	
Excision Lesion/Tumour	30021	Excision Cervical Polyp	low	
Excision Lesion/Tumour	90012	Excision Cyst/Lesion - Facial Bone	low	
Excision Gynecomastia	80010	Excision Gynecomastia	low	
Excision Lesion/Tumour	50021	Excision Lesion Tongue	low	
Excision Lesion/Tumour	20140	Excision Lesion/Tumor - Bone	moderate	
Excision Lesion/Tumour	20139	Excision Lesion/Tumor - Mucous Membrane	low	
Excision Lesion/Tumour	20141	Excision Lesion/Tumor - Muscle/Tendon/Ligament	low	
Skin Tumour Removal	20137	Excision Lesion/Tumor - Skin	low	
Excision Lesion/Tumour	20142	Excision Lesion/Tumor - Spinal Cord	moderate	
Excision Lesion/Tumour	20138	Excision Lesion/Tumor - Visceral	low	
Nerve Surgery	10084	Excision Neuroma	low	
Excision Lesion/Tumour	20029	Excision Pelvic Mass	moderate	
Excision Lesion/Tumour	50022	Excision Salivary Gland	low	
Oral Cavity & Pharynx Surgery	90026	Excision Soft Tissue Mouth	low	
Skin Surgery	20143	Excision Sweat Glands	low	
Cyst/Ganglion Removal	50024	Excision Thyroglossal Cyst	low	
Excision Lesion/Tumour	20039	Excision Warts	low	
Esophagectomy	20036	Excision Zenker'S Diverticulum	moderate	
Other Orthopaedic Surgery	10151	Excision/Debridement Bone	moderate	
Examination Under Anaesthetic	20037	Exploration Common Bile Duct	moderate	
Examination Under Anaesthetic	50025	Exploration Neck	moderate	
Nerve Surgery	70011	Exploration Nerve	low	
Other Eye Surgery	11015	Eyelid Repair	low	
All Other Procedures	80096	Face Sling Repair	low	
Facial Bone Reconstruction	90004	Facial Bone Osteotomy/Reconstruction	low	
Other Orthopaedic Surgery	10085	Fasciotomy/Fasciectomy/Myotomy/Release/Tenolysis	low	
Skin Surgery	80022	Forehead/Brow Lift/Face Lift	low	
Hernia Repair - Hiatal	20042	Fundoplication	moderate	
All Other Procedures	20046	Gastrectomy	moderate	
All Other Procedures	20134	Gastrointestinal bypass	moderate	
Bariatric Surgery	20135	Gastrointestinal Bypass - Bariatric	moderate	
All Other Procedures	80101	Gender Affirmation Metoidioplasty	moderate	
All Other Procedures	80104	Gender Affirmation Phallo/Meta Revision	moderate	
All Other Procedures	80103	Gender Affirmation Phallo/Meta Staging	moderate	
All Other Procedures	80100	Gender Affirmation Phalloplasty	moderate	

Provincial Procedure Group Name	Provincial Procedure Code	Provincial Procedure Name	Surgical Invasiveness Ranking	Low Risk Urologic Surgery Needing Preop GFR Na K
All Other Procedures	80102	Gender Affirmation Vaginoplasty	moderate	
All Other Procedures	80105	Gender Affirmation Vaginoplasty Revision	moderate	
Oral Cavity & Pharynx Surgery	90015	Gingivectomy/Soft Tissue Recontouring	low	
Excision Lesion/Tumour	50026	Glossectomy	moderate	
All Other Procedures	10163	Harvest/Procurement/Biopsy Bone Marrow	low	
Not Reported	12014	Heart Transplant	high	
All Other Procedures	20150	Hemorrhoid Banding/ Ligation	low	
Rectal Surgery	20103	Hemorrhoidectomy	low	
Other Urology Surgery	60023	Hydrocelectomy	low	
Uterine Surgery	30026	Hymenectomy	low	
Excision Lesion/Tumour	70015	Hypophysectomy Transsphenoidal	moderate	
Uterine Surgery	30027	Hysterectomy	moderate	
Uterine Surgery	30077	Hysterectomy And Repair	moderate	
Uterine Surgery	30028	Hysteroplasty	moderate	
Uterine Surgery	30029	Hysteroscopy	low	
Other Urology Surgery	60024	Ileoconduit	moderate	
Colostomy/Ileostomy	20053	Ileostomy	moderate	
Wound/Laceration Care	20132	Incision And Drainage	low	
Oral Cavity & Pharynx Surgery	50027	Incision And Drainage Tonsillar Abscess	moderate	
All Other Procedures	20117	Injection/Installation/Insertion Therapeutic Agent	low	
Vascular Surgery - Other	40017	Insert Vena-Caval Filter	moderate	
Other Eye Surgery	11046	Insertion Orbital Implant	low	
Not Reported	40018	Insertion Pacemaker/Defibrillators - Includes Icd/Aicd	low	
Skin Surgery	80087	Insertion Tissue Expander	low	
Other Urology Surgery	60026	Insertion/Removal Artificial Urinary Sphincter	low	Y
All Other Procedures	20121	Insertion/Removal Catheter Tube	moderate	
All Other Procedures	100009	Insertion/Removal Chest Tube	low	
CSF Drainage	70021	Insertion/Removal Csf Drainage	low	
Cranial Surgery	70016	Insertion/Removal Intracranial Pressure Monitor	low	
All Other Procedures	40040	Insertion/Removal Loop Recorder	low	
All Other Procedures	30031	Insertion/Removal Of Iud	low	
Shoulder Surgery	20120	Insertion/Removal Of Shunt	low	
Wound/Laceration Care	20083	Insertion/Removal Packing	low	
Male Reproductive Surgery	60027	Insertion/Removal Penile Prosthesis	low	
All Other Procedures	70045	Insertion/Removal Peripheral Nerve Stimulator	low	
All Other Procedures	20085	Insertion/Removal Peritoneal Catheter	low	
Vascular Surgery - Other	40043	Insertion/Removal Renal Vascular Access Device	moderate	
Spinal/Back Surgery	70019	Insertion/Removal Spinal Infusion Pump	low	
All Other Procedures	100011	Insertion/Removal Stent - Esophageal	low	
All Other Procedures	20118	Insertion/Removal Stents	moderate	
Wound/Laceration Care	80054	Insertion/Removal Suture	low	
Male Reproductive Surgery	60028	Insertion/Removal Testicular Prosthesis	low	
Vascular Surgery - Other	20061	Insertion/Removal Venous Access Device	low	

Provincial Procedure Group Name	Provincial Procedure Code	Provincial Procedure Name	Surgical Invasiveness Ranking	Low Risk Urologic Surgery Needing Preop GFR Na K
Cranial Surgery	70023	Insertion/Removal/Internalization Of Deep Brain Implant	low	
Spinal/Back Surgery	70018	Insertion/Removal/Internalization Spinal Cord Implant	low	
Other Eye Surgery	11017	Iridectomy	low	
Other Eye Surgery	11040	Iridoplasty	low	
Wound/Laceration Care	20125	Irrigation & Debridement Wound/Soft Tissue	low	
Wound/Laceration Care	20125	Irrigation & Debridement Wound/Soft Tissue	moderate	
Knee - ACL Repair	10035	Knee Ligament Repair	low	
Spinal/Back Surgery	70024	Kyphoplasty	low	
Spinal/Back Surgery	70025	Laminectomy	moderate	
Spinal/Back Surgery	70043	Laminotomy	moderate	
Laparoscopy	30033	Laparoscopy	low	
Examination Under Anaesthetic	30072	Laparoscopy And Hysteroscopy	low	
Laparotomy	20063	Laparotomy	moderate	
All Other Procedures	50028	Laryngectomy And Pharyngectomy	moderate	
Not Reported	50029	Laryngoscopy With Or Without Biopsy	moderate	
Tendon Surgery	10090	Lengthening Tendon	low	
Varicose Veins Ligation/Stripping	40020	Ligation And Stripping Varicose Veins Includes Sclerotherapy	low	
All Other Procedures	40042	Ligation Esophageal Varicies	low	
Vascular Surgery - Other	40044	Ligation Of Artery/Vein	moderate	
Vascular Surgery - Other	40021	Ligation/Repair Av/Vascular Fistula	moderate	
Lipectomy	80033	Lipectomy/Liposuction	low	
Skin Surgery	80097	Lower Body Lift	moderate	
CSF Drainage	70030	Lumbar Puncture	low	
Lung Surgery	50059	Lung Lavage Total	low	
All Other Procedures	60017	Lymphadenectomy	low	
All Other Procedures	30037	Lysis Of Adhesions	moderate	
Not Reported	12011	Major Non-Ohs	moderate	
Breast Reconstruction	80034	Mammoplasty Augmentation	low	
Breast Reduction	80035	Mammoplasty Reduction	low	
Other Orthopaedic Surgery	10091	Manipulation Joint	low	
D&C and Related Surgery	30038	Manual Removal Placenta	low	
Mastectomy	20065	Mastectomy	low	
All Other Procedures	80098	Mastectomy - Transgender	low	
Mastoidectomy	50031	Mastoidectomy	low	
Breast Reconstruction	80040	Mastopexy	low	
Other Urology Surgery	60032	Meatoplasty - Urethral	low	
Other Ear Surgery	20119	Meatoplasty/Canaloplasty (Ear)	low	
Other Urology Surgery	60033	Meatotomy - Urethral	low	
Scope of Chest	100013	Mediastinoscopy	moderate	
Scope of Chest	100014	Mediastinotomy	moderate	
Knee - Meniscectomy	10092	Meniscetomy/Meniscal Repair Knee	low	
Not Reported	12012	Minor Non-Ohs	moderate	
Other Orthopaedic Surgery	20128	Myotomy	low	

Provincial Procedure Group Name	Provincial Procedure Code	Provincial Procedure Name	Surgical Invasiveness Ranking	Low Risk Urologic Surgery Needing Preop GFR Na K
Myringotomy	50032	Myringotomy With Or Without Insertion Of Tubes	low	
Nasal Surgery	50033	Nasal Polypectomy	low	
Other Urology Surgery	60034	Nephrectomy	moderate	
Kidney/Bladder Stone Removal	60036	Nephro/Pyelo/Ureteral/Urethral/Lithotripsy	low	Y
Kidney/Bladder Stone Removal	60075	Nephro/Pyelo/Ureteral/Urethrolithotomy	low	Y
Other Urology Surgery	60076	Nephroscopy	low	Y
Other Urology Surgery	60037	Nephrotomy/Nephrostomy	moderate	
Other Urology Surgery	60038	Nephroureterectomy	moderate	
Nerve Surgery	70039	Nerve Block/Pain Control	low	
Male Reproductive Surgery	60039	Nesbit Procedure	low	
Nerve Surgery	70032	Neurectomy	low	
Nerve Surgery	10093	Neurolysis	low	
Breast Reconstruction	80042	Nipple Reconstruction	low	
Not Reported	20149	Non-Surgical Code	low	
Excision Lesion/Tumour	20066	Omentectomy	low	
Fallopian Tube/Ovarian Surgery	30040	Oophorectomy	low	
Fallopian Tube/Ovarian Surgery	30074	Oophoropexy	low	
Bladder Surgery	60040	Operations For Urinary Incontinence	low	
Other Eye Surgery	11021	Orbital Decompression	low	
Other Eye Surgery	11024	Orbitotomy - Anterior (For Lacrimal Gland)	low	
Other Eye Surgery	11025	Orbitotomy - Posterior (For Orbital Apex, Optic Nerve)	low	
Male Reproductive Surgery	60041	Orchiectomy	low	
Male Reproductive Surgery	60042	Orchiopexy	low	
All Other Procedures	20123	Organ Retrieval	moderate	
All Other Procedures	20124	Organ Transplant	high	
Oral Cavity & Pharynx Surgery	90024	Orthodontic Procedures	low	
Other Orthopaedic Surgery	10154	Osteotomy/Repair Bone Or Joint Not Classified Elsewhere	low	
Not Reported	12009	Other Cardiac Surgery	high	
Other Ear Surgery	50063	Other Ear Implant	moderate	
All Other Procedures	20146	Other General Surgery	moderate	
All Other Procedures	30076	Other Gynecology Surgery	moderate	
All Other Procedures	70044	Other Neurosurgery	moderate	
Not Reported	12007	Other Ohs	high	
Other Eye Surgery	11047	Other Ophthalmology Surgery	low	
Oral Cavity & Pharynx Surgery	90025	Other Oral Maxillofacial Surgery	low	
Other Orthopaedic Surgery	10160	Other Orthopedic Surgery	moderate	
All Other Procedures	50060	Other Otolaryngology Surgery	low	
All Other Procedures	80095	Other Plastic Surgery	moderate	
All Other Procedures	100033	Other Thoracic Surgery	moderate	
Other Urology Surgery	60078	Other Urology Surgery	low	
Vascular Surgery - Other	40046	Other Vascular Surgery	moderate	
Other Ear Surgery	80048	Otoplasty	low	
Fallopian Tube/Ovarian Surgery	30041	Ovarian Cystectomy	low	

Provincial Procedure Group Name	Provincial Procedure Code	Provincial Procedure Name	Surgical Invasiveness Ranking	Low Risk Urologic Surgery Needing Preop GFR Na K
Fallopian Tube/Ovarian Surgery	30042	Ovarian Drilling/Wedge Resection	low	
All Other Procedures	20068	Pancreatectomy	moderate	
All Other Procedures	20069	Pancreaticoduodenectomy	high	
All Other Procedures	20067	Pancreaticojejunostomy	moderate	
All Other Procedures	50036	Parathyroidectomy	low	
Parotidectomy	50037	Parotidectomy	low	
Other Joint Reconstruction	10126	Patella Realignment	low	
Other Orthopaedic Surgery	10127	Patellectomy	low	
All Other Procedures	80085	Pedicle Flap Graft	low	
All Other Procedures	30073	Pelvic/Abdominal Exenteration	moderate	
Male Reproductive Surgery	60043	Penectomy	low	
Male Reproductive Surgery	60044	Penis Reconstruction	low	
All Other Procedures	100015	Pericardial Window	moderate	
All Other Procedures	30079	Perinatal Intervention	low	
Vaginal Repair	30044	Perineorrhaphy	low	
Oral Cavity & Pharynx Surgery	80049	Pharyngoplasty	low	
Lung Surgery	100016	Pleurectomy	moderate	
Scope of Chest	100017	Pleuroscopy	low	
Lung Surgery	100018	Pneumonectomy	high	
Lens & Vitreous (non-cataract) Surgery	11026	Posterior Vitrectomy/Gas Exchange/Membrane Peel	low	
Prostate Surgery	60045	Prostatectomy	moderate	
Other Eye Surgery	11027	Pterygium Excision	low	
Other Eye Surgery	11028	Ptosis Repair	low	
Lung Surgery	100031	Pulmonary Lobectomy	moderate	
Not Reported	12004	Pure Cabg	high	
Not Reported	12006	Pure Valve	high	
Other Urology Surgery	60047	Pyeloplasty	moderate	
All Other Procedures	20074	Pyloroplasty	moderate	
All Other Procedures	80092	Reattachment Of Limb (Except Thumb And Fingers)	moderate	
Other Eye Surgery	11029	Recession/Resection/Myotomy	low	
Other Ear Surgery	50039	Reconstruction Auricle	low	
Breast Reconstruction	80050	Reconstruction Breast; Incl: Mastectomy With Immed Reconstruction. If	low	
Tonsillectomy/Adenoidectomy	50040	Reconstruction Ossicle	low	
Fracture Repair	10148	Reduction With Or Without External Fixation	low	
Other Urology Surgery	60050	Reimplantation Ureteral	moderate	
All Other Procedures	20075	Release Bowel Obstruction	moderate	
Hand/Wrist Surgery	20076	Release Carpal Tunnel	low	
All Other Procedures	80088	Release Syndactlyl/Webb	low	
Nerve Surgery	20077	Release Tarsal Tunnel	low	
Oral Cavity & Pharynx Surgery	50041	Release Tongue Tie	low	
Cyst/Ganglion Removal	30046	Removal Bartholin Cyst	low	
Breast Reconstruction	80053	Removal Breast Implant	low	
Foreign Body Removal Surgery	20081	Removal Foreign/Loose Bodies	low	

Provincial Procedure Group Name	Provincial Procedure Code	Provincial Procedure Name	Surgical Invasiveness Ranking	Low Risk Urologic Surgery Needing Preop GFR Na K
Other Orthopaedic Surgery	10135	Removal Hardware	low	
Foreign Body Removal Surgery	11030	Removal Intraocular Fbody	low	
Myringotomy	50042	Removal Myringotomy Tube(S)	low	
All Other Procedures	20144	Removal Prosthesis Non-Orthopedic	low	
Skin Surgery	80055	Removal Tissue Expander/Implant	low	
Other Orthopaedic Surgery	20082	Removal Toenail/Fingernail	low	
Vascular Surgery - Other	40039	Repair Aneurysm - Non Aortic	moderate	
Aortic Aneurysm Repair	40026	Repair Aneurysm Aortic	high	
Bladder Surgery	60054	Repair Bladder Laceration	low	
Other Eye Surgery	11043	Repair Chalazion/Conjunctiva	low	
Hernia Repair - Chest Wall	100020	Repair Chest Wall	moderate	
Oral Cavity & Pharynx Surgery	80056	Repair Cleft Lip	low	
Oral Cavity & Pharynx Surgery	80057	Repair Cleft Palate	low	
Vaginal Repair	30078	Repair Cystocele/Enterocele /Rectocele/Repair Pelvic Floor	low	
All Other Procedures	100019	Repair Diaphragm	moderate	
Spinal/Back Surgery	70042	Repair Dura	moderate	
Other Eye Surgery	11042	Repair Eye Trauma - Laceration	low	
Fracture Repair	10152	Repair Fracture	low	
Fracture Repair	10104	Repair Fracture Hip	moderate	
All Other Procedures	20013	Repair Gastrointestinal Ulcer	moderate	
Hernia Repair - Abdominal	20087	Repair Hernia Abdominal Diaphramatic	moderate	
Hernia Repair - Abdominal	20088	Repair Hernia Epigastric	low	
Hernia Repair - Abdominal	20089	Repair Hernia Femoral	low	
Hernia Repair - Hiatal	20090	Repair Hernia Hiatal	moderate	
Hernia Repair - Abdominal	20091	Repair Hernia Incisional	moderate	
Hernia Repair - Abdominal	20092	Repair Hernia Inguinal	low	
Hernia Repair - Abdominal	20093	Repair Hernia Paracolostomy	moderate	
Hernia Repair - Abdominal	20094	Repair Hernia Umbilical	low	
Hernia Repair - Abdominal	20095	Repair Hernia Ventral	moderate	
Other Urology Surgery	60055	Repair Hypospadias	low	Y
Wound/Laceration Care	11032	Repair Laceration - Non Ophthalmology	low	
Ligament Surgery	80082	Repair Ligament	low	
All Other Procedures	10155	Repair Muscle	low	
Spinal/Back Surgery	70033	Repair Myelomeningocele	moderate	
Nerve Surgery	80089	Repair Nerve	low	
Fistula Repair - Non Vascular	20126	Repair Non-Vascular Fistula	moderate	
Vascular Surgery - Other	40032	Repair Of Artery/Vein	moderate	
All Other Procedures	100021	Repair Pectus Carinatum/Excavatum	moderate	
Rectal Surgery	20071	Repair Rectal Prolapse	low	
Shoulder Surgery	10156	Repair Shoulder/Rotator Cuff	low	
Tendon Surgery	80083	Repair Tendon	low	
Male Reproductive Surgery	60079	Repair Torsion Of Testicle	low	
All Other Procedures	80064	Repair Trauma Hand/Fingers - Includes Reattachment	low	

Provincial Procedure Group Name	Provincial Procedure Code	Provincial Procedure Name	Surgical Invasiveness Ranking	Low Risk Urologic Surgery Needing Preop GFR Na K
Wound/Laceration Care	20096	Repair Wound Dehiscence	low	
Not Reported	40029	Reposition/Replace/Remove Pacemaker/Defibrillator/Icd/Aicd Wire/Lead	low	
Excision Lesion/Tumour	50061	Resection Composite Neck	moderate	
Bowel Resection	20098	Resection Large Bowel	moderate	
Excision Lesion/Tumour	20099	Resection Liver	moderate	
Excision Lesion/Tumour	20100	Resection Mediastinum	moderate	
Rib Resection	100025	Resection Rib	moderate	
Bowel Resection	20101	Resection Small Bowel	moderate	
All Other Procedures	50058	Resection Trachea Includes Tracheo-Esophageal Puncture	moderate	
All Other Procedures	100026	Resection Xiphi-Sternum	moderate	
All Other Procedures	20148	Resection/Reconstruction Bile Duct	moderate	
Other Eye Surgery	11034	Retinal Detachment Scleral Buckle	low	
Skin Surgery	80067	Revision Scar	low	
Nasal Surgery	80068	Rhinoplasty	low	
Oral Cavity & Pharynx Surgery	50057	Rhinotomy/Maxillectomy	low	
Nerve Surgery	70034	Rhizotomy	low	
Fallopian Tube/Ovarian Surgery	30055	Salpingectomy	low	
Fallopian Tube/Ovarian Surgery	30058	Salpingo-Oophorectomy	low	
Fallopian Tube/Ovarian Surgery	30057	Salpingolysis	low	
Fallopian Tube/Ovarian Surgery	30059	Salpingoplasty	low	
Fallopian Tube/Ovarian Surgery	30060	Salpingostomy	low	
Nasal Surgery	50043	Septoplasty	low	
Nasal Surgery	50044	Septorhinoplasty	low	
Excision Lesion/Tumour	50054	Sialolithotomy	low	
Sinus Surgery	50045	Sinus Irrigation	low	
Skin Surgery	80090	Skin Graft	low	
Male Reproductive Surgery	60059	Spermatocelectomy	low	
Sphincterotomy	20106	Sphincterotomy	low	
Spinal/Back Surgery	70040	Spinal Fusion	moderate	
All Other Procedures	20107	Splenectomy	moderate	
Tympanoplasty	50046	Stapedectomy	low	
Thoracotomy	100027	Sternotomy With Or Without Excision Mediastinal Mass	moderate	
Nerve Surgery	70031	Sympathectomy	low	
Other Joint Reconstruction	10141	Synovectomy/Debridement Joint	low	
Other Eye Surgery	11035	Tarsorrhaphy	low	
Biopsy in OR	11036	Temporal Artery Biopsy	low	
Other Orthopaedic Surgery	10142	Tenolysis	low	
Tendon Surgery	10143	Tenotomy	low	
Scope of Chest	100028	Thoracoscopy	low	
Thoracotomy	100029	Thoracotomy	high	
Vascular Surgery - Other	40038	Thrombectomy/Embolectomy	moderate	
All Other Procedures	100030	Thymectomy	moderate	
Thyroidectomy	50047	Thyroidectomy	low	

Provincial Procedure Group Name	Provincial Procedure Code	Provincial Procedure Name	Surgical Invasiveness Ranking	Low Risk Urologic Surgery Needing Preop GFR Na K
All Other Procedures	50048	Thyroplasty	low	
Tonsillectomy/Adenoidectomy	50049	Tonsillectomy	low	
Dental Surgery	90016	Tooth Extraction	low	
Other Eye Surgery	11037	Trabeculectomy	low	
Cataract Surgery	11049	Trabeculectomy & Cataract Extraction & Iol Implant	low	
All Other Procedures	50050	Tracheostomy	low	
Excision Lesion/Tumour	20131	Transanal Excision Rectal Tumor/Lesion	low	
Nerve Surgery	70035	Transposition Nerve	low	
Bladder Surgery	60060	Transurethral Resection Bladder Tumor	moderate	
Bladder Surgery	60061	Transurethral Resection Of Bladder Neck	low	Y
Prostate Surgery	60062	Transurethral Resection Prostate	moderate	
Fallopian Tube/Ovarian Surgery	30063	Tubal Ligation	low	
Fallopian Tube/Ovarian Surgery	30034	Tubal Patency Testing	low	
Fallopian Tube/Ovarian Surgery	30064	Tubal Reanastomosis	low	
Excision Lesion/Tumour	11038	Tumour Excision Eye	low	
Nasal Surgery	50051	Turbinectomies	low	
Tympanoplasty	50052	Tympanoplasty	low	
Spinal/Back Surgery	70036	Untethering Spinal Cord	moderate	
Skin Surgery	80002	Upper Body Lift	moderate	
Other Urology Surgery	60063	Ureterectomy	moderate	
Other Urology Surgery	60065	Ureteroplasty	moderate	
Other Urology Surgery	60066	Ureteroscopy	low	Y
Other Urology Surgery	60067	Ureterostomy	low	
Other Urology Surgery	60077	Urethrectomy	low	Y
Other Urology Surgery	60069	Urethroplasty	low	
Other Urology Surgery	60070	Urethrotomy	low	Y
Uterine Surgery	30039	Uterine Myomectomy/Fibroidectomy	moderate	
Uterine Surgery	30065	Uterine Suspension	low	
All Other Procedures	50053	Uvulopalatopharyngoplasty/Laryngoplasty	moderate	
All Other Procedures	30066	Vaginal Delivery	low	
Nerve Surgery	20110	Vagotomy	low	
Not Reported	12005	Valve + Other	high	
Male Reproductive Surgery	60071	Varicocelectomy	low	
Vascular Bypass Graft - Non Cardiac	40037	Vascular Bypass Graft - Non Cardiac Includes Removal	moderate	
Male Reproductive Surgery	60072	Vasectomy	low	
Male Reproductive Surgery	60074	Vasovasostomy/Vasoepididymostomy	low	
Cranial Surgery	70037	Ventriculostomy	low	
Spinal/Back Surgery	70038	Vertebrectomy	moderate	
Oral Cavity & Pharynx Surgery	90007	Vestibuloplasty	low	
Vaginal Repair	30067	Vulvectomy	moderate	
Vaginal Repair	30068	Vulvoplasty/Vaginoplasty	low	